MIL-L-81174/2B
21 September 1972
SUPERSEDING
MIL-L-81174/2
7 December 1964

MIL-L-0081174/2A(AS)
20 January 1968

MILITARY SPECIFICATION SHEET

LIGHT, LANDING, AIRCRAFT, CONTROLLABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

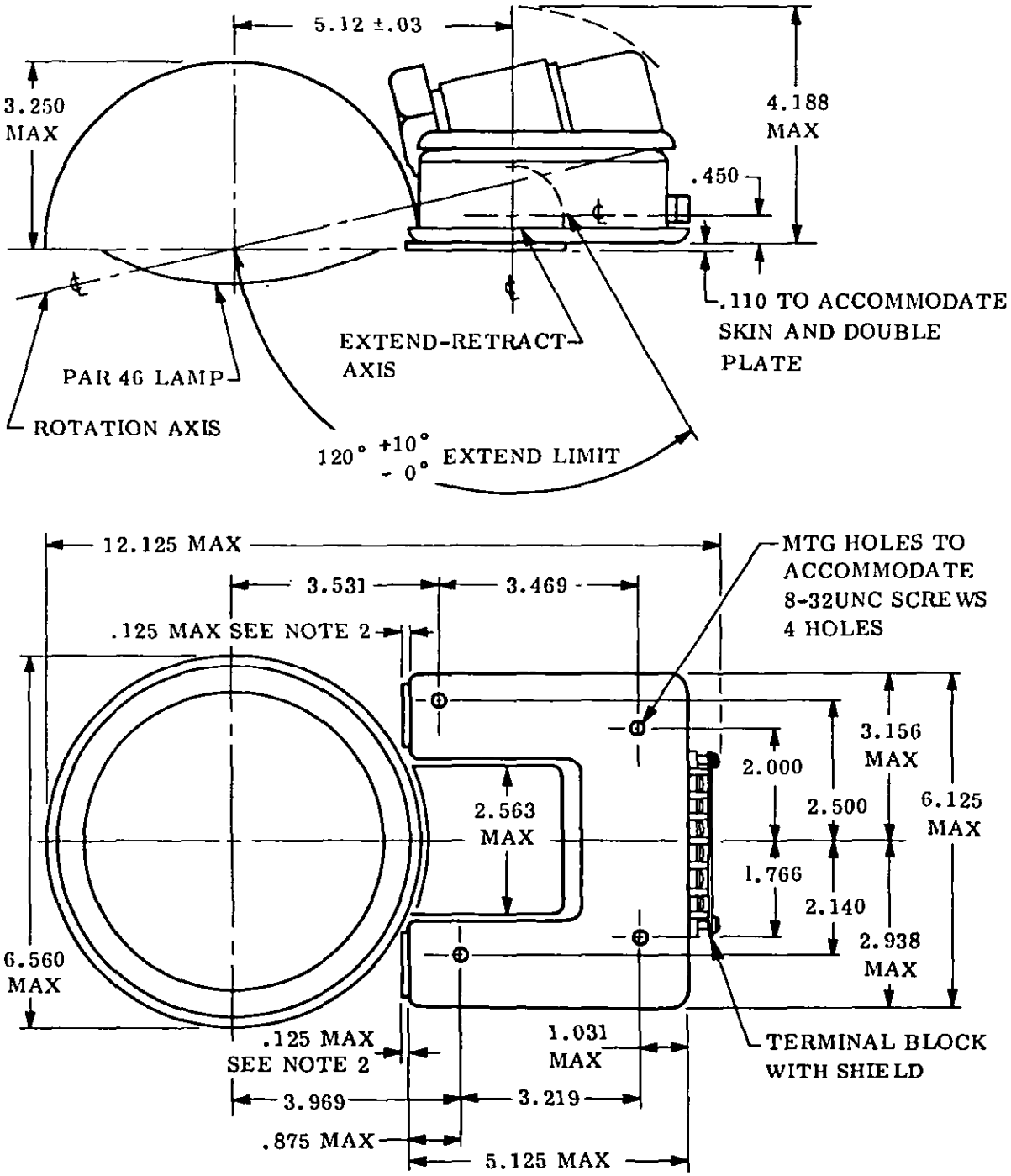
The complete requirements for procuring the light described herein shall consist of this document and the issue in effect of MIL-L-81174.

TABLE I

Dash Number	
-1	with cam operated limit switch (See Fig 2)
-1A	with cam operated limit switch (See Fig 2 and "Vibration" below)
-2	without cam operated limit switch (See Fig 3)
-2A	without cam operated limit switch (See Fig 3 and "Vibration" below)

FSC 6220

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

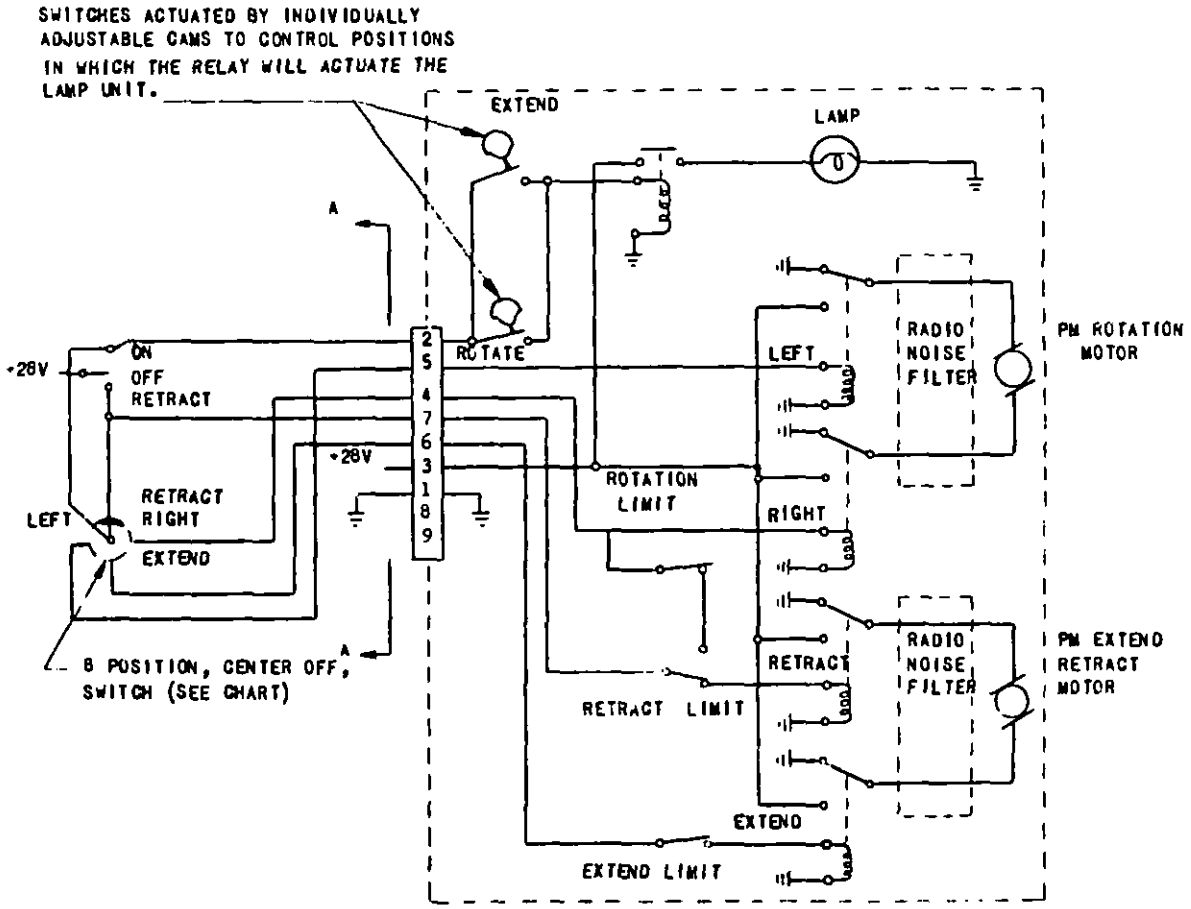


NOTES:

1. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.031.
2. MAXIMUM POSITION OF RELAY BOX AFT OF MOUNTING PLATE. PREFERRED CONSTRUCTION SHALL NOT EXCEED 5.125 DIMENSION.

Figure 1 - Controllable Aircraft Landing Light

Source: https://assist.dla.mil -- Downloaded: 20260-23T05:03Z
Check the source to verify that this is the current version before use.



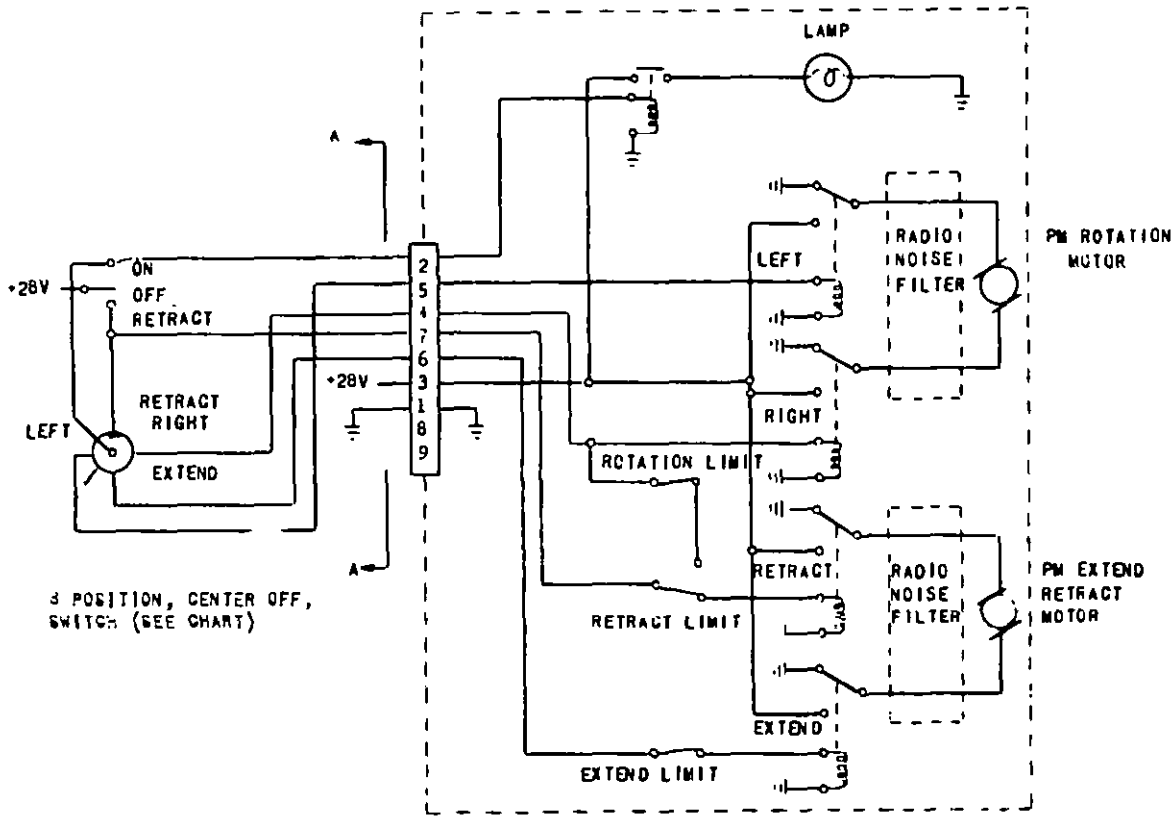
SWITCH POSITIONS
RETRACT
RETRACT AND RIGHT
RIGHT
RIGHT AND EXTEND
EXTEND
EXTEND AND LEFT
LEFT
LEFT AND EXTEND

NOTES:

1. POSITION OF LIMIT SWITCHES INDICATE LIGHT UNIT PARTIALLY EXTENDED AND ROTATED.
2. ONLY PORTION OF FIGURE TO RIGHT OF A-A IS FURNISHED AS PART OF LIGHT ASSEMBLY. THAT PART TO LEFT OF A-A IS PART OF AIRCRAFT CIRCUIT.

Figure 2 - Type MA-3, Part No. M81174/2-1, Circuit Diagram

Source: https://assist.dla.mil -- Downloaded: 202640-23T05:03Z
Check the source to verify that this is the current version before use.



SWITCH POSITIONS
RETRACT
RETRACT AND RIGHT
RIGHT
RIGHT AND EXTEND
EXTEND
EXTEND AND LEFT
LEFT
LEFT AND RETRACT

NOTES:

1. POSITIONS OF LIMIT SWITCHES INDICATE LIGHT UNIT PARTIALLY EXTENDED AND ROTATED.
2. ONLY PORTION OF FIGURE TO RIGHT OF A-A IS FURNISHED AS PART OF LIGHT ASSEMBLY. THAT PART TO LEFT OF A-A IS PART OF AIRCRAFT CIRCUIT.

Figure 3 - Part No. M81174/2-2 Circuit Diagram

Source: https://assist.dla.mil -- Downloaded: 20260-23T05:03Z
Check the source to verify that this is the current version before use.

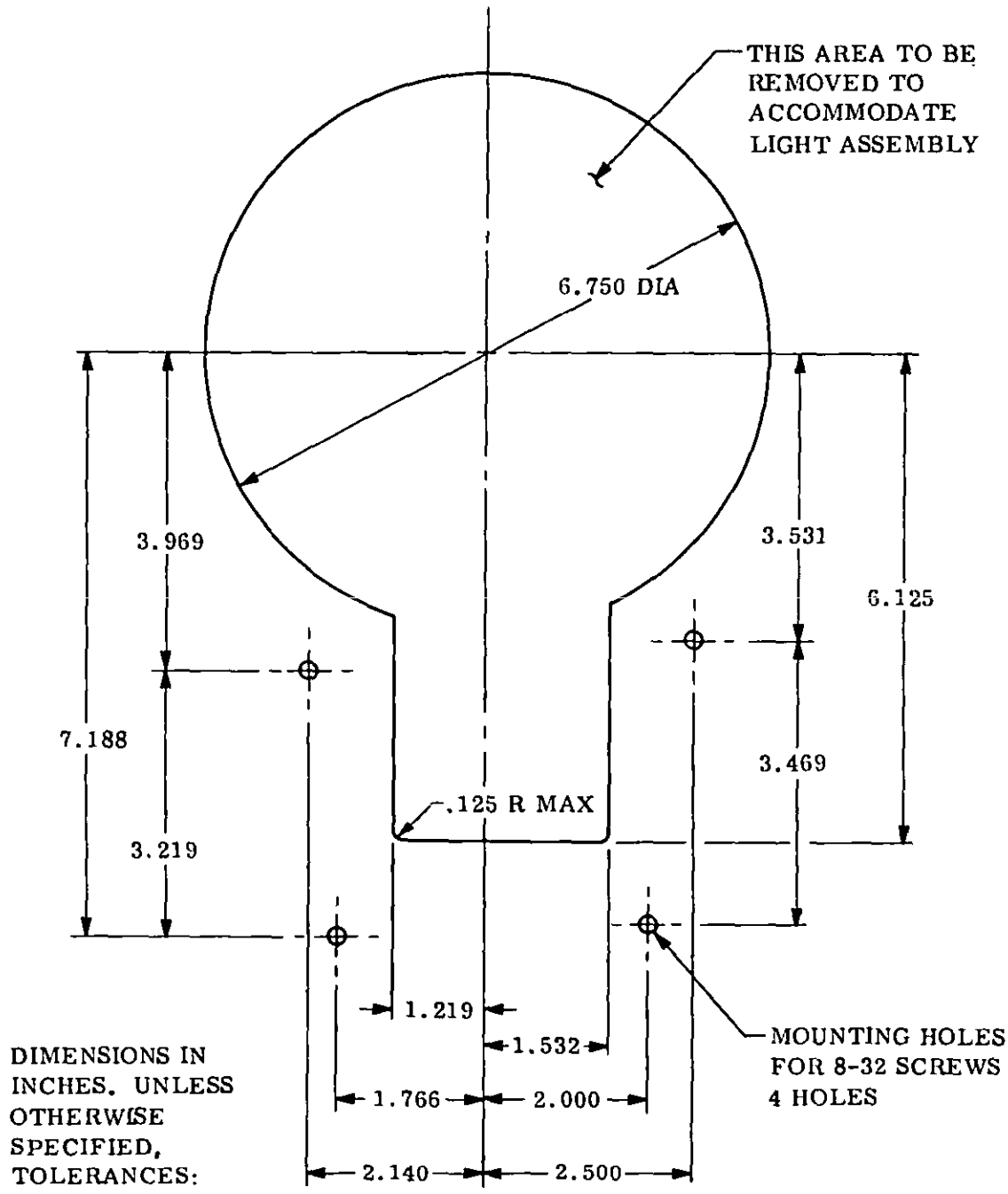


FIGURE 4 - DESIGN FOR INSTALLATION AREA

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See Figures 1 and 3.

Extending limit: Fully extended position shall be not less than 120° nor more than 130° from the mounting plane of the light.

Braking: Within 0.5° under an air load of 10 pounds whether extending, retracting, or rotating.

Lamp: Light shall accommodate a PAR 46, 450 watt screw terminal base lamp, ASA No. 4580 or Q4681.

Lamp switching device: A relay or solid state switching device, shall be provided for controlling the current to the lamp and shall be an integral part of the light.

Rotation: The light shall be capable of rotating in both clockwise and counterclockwise directions, and stopping the light by means of an electric actuating mechanism, controlled by a remotely situated, eight-position switch.

Wiring: See Figure 2 for type MA-3, Part No. M81174/2-1 and Figure 3 for Part No. M81174/2-2.

Weight: 6 pounds maximum including lamp.

Extension time from fully retracted position to fully extended position shall be 10 seconds maximum with a 10 pound air load applied.

Stowage time from fully extended position to fully retracted position shall be 10 seconds maximum with a 10 pound air load applied.

Vibration: For -1A and -2A lights only, the vibratory acceleration applied from 52 to 500Hz shall be ±5g instead of ±10g. The lamp shall be on during the vibration tests and cycled through all switch positions sufficiently to simulate the conditions anticipated during actual use of the light.

Radio interference: Radio interference tests shall be conducted in accordance with MIL-STD-461. The light shall be controlled by a MS 28939-1 type switch. However, the switch may be isolated from the test enclosure (screen room) during the measurements.

Lamp cut-off cams: For -1 and -1A lights the cams of the lights shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

REQUIREMENTS (Continued):

factory adjusted so that the lamp will not be energized during a rotated position of 90° left to 90° right when the light is in a position of fully retracted to 60° extended.

Rotation time 360° in both clockwise and counterclockwise directions shall be 15 seconds maximum and 10 seconds minimum (10-lb load not applied)

Part number: Consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

M81174/2-1

Basic document number

Dash number from table I

Supersession data: This specification sheet in conjunction with MIL-L-81174 supersedes MIL-L-26998.

Custodians:

Navy - AS
Air Force - 82
Army - AV

Preparing Activity:

Navy - AS

Project No. 6220-0206

Review Activities:

Navy -
Air Force -
Army -

User Activities

Navy - MC, CG
Air Force -
Army -

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:03Z
Check the source to verify that this is the current version before use.